Vehicle Tracking + Occlusions Research Project

Phase 1H Research Documentation

Identity Matching, Memory, Ambiguity, and Reliability-Aware Selective Routing

---

Purpose: archival documentation of the Phase 1H design decisions, controlled alternatives, corrections, diagnostics, results, limitations, and freeze decision.

Phase status:	Complete and frozen
Dataset used:	UA-DETRAC frozen pilot events
Eligible identity events:	10 recorded-reappearance events (E001–E009 except E010, plus E011)
Input dependency:	Phase 1G filtered road-vehicle candidate set
Primary appearance encoder:	ImageNet-pretrained ResNet50, 2048-D L2-normalized embedding
Key phase output:	Reliability-aware decision routing: MATCH or NEED MORE EVIDENCE
Next phase:	Phase 1I systematic failure taxonomy / error analysis

Documentation generated from the final Phase 1H experimental record. Provisional or invalid intermediate results are retained only when they explain a design correction; frozen conclusions are explicitly identified.

Document Map

1. Phase objective and research boundary
2. System position and experimental contract
3. Implementation artifacts
4. H-0 baseline identity matcher
5. H-1 M1 positive target memory
6. H-1 M2 distractor-aware memory
7. M2 failure mechanism and redesign lesson
8. Ambiguity signals from candidate competition
9. E004 confidently-wrong failure analysis
10. H-2 candidate-overlap / duplicate diagnostic
11. H-3 observation-quality diagnostics
12. H-3 R1 versus exploratory R2 selective routing
13. Consolidated Phase 1H findings
14. Scientific cautions and limitations
15. Phase 1H freeze decision and handoff to Phase 1I

1 Phase Objective and Research Boundary

Phase 1H was designed to answer the identity-selection question that Phase 1G deliberately left open: *once the returned target is contained in a plausible road-vehicle candidate set, can appearance memory, distractor awareness, ambiguity analysis, and reliability-aware routing reduce false identity decisions?*

The user still performs only the initial target selection. Phase 1H is automatic. It receives candidate vehicles from the frozen Phase 1G generator and reasons about identity. The phase intentionally does **not** yet implement the final sequential acquisition of plate, color, make/model, local features, or other cues. Instead, it establishes the appearance-based identity baseline and determines when appearance evidence should be trusted versus deferred.

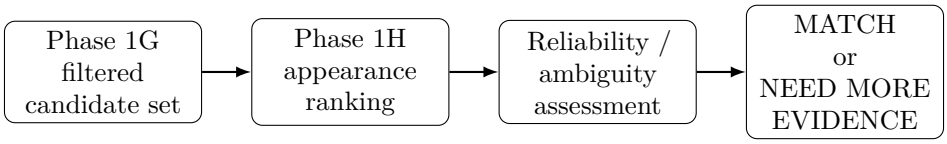


Figure 1: Position of Phase 1H. Phase 1G proposes plausible vehicles; Phase 1H ranks identity evidence and decides whether that evidence is reliable enough to accept.

Frozen finding. Phase 1H changes the project from "always choose a best-looking candidate" to "choose only when evidence is sufficiently reliable; otherwise defer and request more evidence."

2 System Position and Experimental Contract

The frozen Phase 1G baseline retained the true target in all 10 eligible recorded-reappearance events. Therefore Phase 1H could separate candidate availability from identity selection. If the target is in the candidate set but is ranked below another vehicle, the failure belongs to identity reasoning (F5), not candidate generation (F4).

The controlled Phase 1H contract was:

- **Candidate source:** frozen Phase 1G real YOLO11n detections after the broad geometric plausibility filter.
- **Identity encoder:** ImageNet-pretrained ResNet50 with classifier removed, producing a 2048-D L2-normalized embedding.
- **Candidate score in H-0/M1:** cosine similarity to the reference/positive target memory.
- **Ground truth use:** only after runtime scoring/ranking, for offline evaluation of target rank and correctness.
- **No additional identity modalities:** no plate OCR, color classifier, make/model classifier, motion identity, or sequential cue policy.

The identity evaluation set consisted of 10 events with a manually verified recorded reappearance: E001, E002, E003, E004, E005, E006, E007, E008, E009, and E011. The no-return case E012 and the non-eligible cases were not forced into identity-reappearance scoring.

3 Implementation Artifacts

The principal Phase 1H implementation/evaluation artifacts are listed below. These paths form the archival implementation record.

- `src/recovery/appearance_encoder.py` - ResNet50 appearance encoder and corrected square-pad preprocessing.
- `src/recovery/recent_memory.py` - M1 recent positive target-memory construction.
- `src/recovery/distractor_memory.py` - M2 runtime hard-distractor memory construction.
- `src/evaluation/test_phase1h0_encoder.py` - encoder/crop sanity check.
- `src/evaluation/test_phase1h0_single_event.py` - E011 single-event H-0 ranking check.
- `src/evaluation/evaluate_phase1h0_all_events.py` - H-0 all-event identity baseline.
- `src/evaluation/test_phase1h1_m1_memory.py` - M1 memory-content audit.
- `src/evaluation/evaluate_phase1h1_m1_all_events.py` - M1 all-event evaluation.
- `src/evaluation/audit_phase1h1_m1_memory_quality.py` - diagnostic tracker-vs-GT memory quality audit.
- `src/evaluation/audit_phase1h1_m2_distractor_opportunities.py` - GT-only diagnostic for hard-distractor opportunities.
- `src/evaluation/audit_phase1h1_m2_runtime_distractors.py` - deployable runtime-only distractor-opportunity audit.
- `src/evaluation/test_phase1h1_m2_memory.py` - negative-memory bank audit.
- `src/evaluation/evaluate_phase1h1_m2_all_events.py` - initial M2 subtraction experiment.

- `src/evaluation/diagnose_phase1h1_m2_failure.py` - M2 over-penalization diagnosis.
- `src/evaluation/diagnose_phase1h1_m2_ambiguity.py` - distractor-memory ambiguity-gap diagnostic.
- `src/evaluation/diagnose_phase1h1_candidate_margin_all.py` - all-event Top1–Top2 margin diagnostic.
- `src/evaluation/visualize_phase1h1_e004_failure.py` - E004 confidently-wrong case visualization.
- `src/evaluation/diagnose_phase1h2_candidate_duplicates.py` - H-2 overlap/duplicate diagnostic.
- `src/evaluation/diagnose_phase1h3_observation_quality.py` - target and Top1 observation-quality audit.
- `src/evaluation/diagnose_phase1h3_runtime_routing_signals.py` - runtime-only routing-signal summary.
- `src/evaluation/diagnose_phase1h3_confidence_consistency.py` - appearance/detector-confidence contradiction diagnostic.
- `src/evaluation/evaluate_phase1h3_selective_routing.py` - R1 vs exploratory R2 selective decision experiment.

Key configuration/results include `configs/phase1/phase1h0_matcher.yaml`, `configs/phase1/phase1h1_m1_memory.yaml`, `configs/phase1/phase1h1_m2_memory.yaml`, and the corresponding `results/phase1/phase1h*.csv` outputs.

4 H-0 Baseline Identity Matcher

4.1 Design

H-0 intentionally used the simplest possible appearance baseline. The user's original selection crop was encoded once. Each Phase 1G candidate at the recorded reappearance frame was encoded with the same ResNet50, and candidates were ranked by cosine similarity to the selection embedding.

A preprocessing correction was required before freezing the encoder. The default ImageNet center-crop pipeline could truncate long, small vehicle crops. The frozen preprocessing therefore became:

1. pad the crop to a square using black background,
2. resize to 224×224 using bilinear interpolation,
3. apply ImageNet normalization,
4. extract and L2-normalize the 2048-D feature vector.

4.2 Single-event sanity check

E011 at frame 378 was used as a difficult taxi-heavy sanity check. The true target was retained by Phase 1G but ranked only 3rd by H-0: target similarity 0.8038 versus strongest distractor similarity 0.8743, giving a negative target-vs-hardest-distractor margin of -0.0706. This immediately demonstrated that candidate generation was not the remaining bottleneck; identity selection was.

4.3 All-event baseline

Table 1: H-0 baseline identity-ranking results across the 10 eligible events.

Metric	Value
Target candidate retained	10/10
Rank-1 accuracy	5/10 = 0.500
Rank-3 accuracy	6/10 = 0.600
Mean target rank	4.70
Mean reciprocal rank (MRR)	0.585
Positive target-vs-hardest margin	5/10
Mean target-vs-hardest margin	+0.009

Event target ranks were: E001=9, E002=1, E003=14, E004=12, E005=1, E006=1, E007=1, E008=4, E009=1, E011=3.

Frozen finding. Phase 1G candidate recall was 10/10, but H-0 Rank-1 identity accuracy was only 5/10. Phase 1H therefore confirmed a genuine identity-selection problem among visually plausible vehicles.

5 H-1 M1: Recent Positive Target Memory

5.1 Initial design and correction

M1 tested whether storing multiple recent views of the target would improve ranking. The intended memory consisted of the original selection plus up to four recent valid SAM2 target crops.

An early implementation incorrectly allowed crops up to the recorded reappearance frame. That made the task artificially easy because frames immediately before reappearance could effectively leak post-occlusion target appearance into memory. The resulting initial 10/10 Rank-1 result was **discarded and never frozen**.

The corrected M1 construction became:

> original selection + up to four most recent valid SAM2 tracker crops strictly before `occlusion_start`, with FIFO capacity 5.

For controlled Phase 1H evaluation, `occlusion_start` is used as an offline experimental cutoff so M1 and M2 receive the same clean pre-occlusion observation window. A future deployable implementation must instead suppress updates using runtime reliability states rather than GT/manual timing.

5.2 Memory quality audit

A GT-only diagnostic checked whether the recent SAM2 target crops were contaminated. Across 26 recent-memory entries:

- mean tracker-vs-GT IoU = 0.7837,
- strong IoU ≥ 0.50: 25/26,
- weak IoU 0.30–0.50: 1/26,
- contaminated IoU < 0.30: 0/26.

This ruled out widespread wrong-target contamination as the explanation for M1's behavior.

5.3 Corrected M1 result

Table 2: H-0 versus corrected M1.

Metric	H-0	M1
Rank-1 accuracy	5/10	5/10
Rank-3 accuracy	6/10	5/10
Mean target rank	4.70	4.40
MRR	0.585	0.574
Positive target margin	5/10	5/10
Mean target margin	+0.009	+0.0222

Relative to H-0, M1 improved E003 from rank 14 to 8, left eight events unchanged, and worsened E011 from rank 3 to 6. The memory bank therefore did not materially improve the overall identity baseline.

Frozen finding. More target views were not automatically better. Even clean target memory can increase the number of ways a visually similar distractor matches one of the stored views.

6 H-1 M2: Distractor-Aware Memory

6.1 Why M2 was justified

The literature review suggested that difficult look-alikes should be remembered deliberately rather than treating all historical observations identically. Before implementing this idea, Phase 1H tested whether the pilot data actually contained hard distractor encounters.

A GT-only diagnostic sampled pre-occlusion frames and defined a provisional hard encounter as appearance similarity ≥ 0.80 and normalized center distance ≤ 3.0. This was a diagnostic definition, not a frozen final policy. Results:

- events with visually similar distractors: 9/10,
- events with similar + nearby (hard) distractors: 7/10,
- total sampled hard rows: 44,
- mean per-event strongest distractor similarity: approximately 0.838.

A separate runtime-only audit then used SAM2 target predictions, YOLO11n vehicle detections, ResNet appearance similarity, spatial proximity, and a geometry reliability gate. It found deployable hard-distractor opportunities in 5/10 events (E003, E005, E008, E009, E011).

6.2 Runtime negative-memory bank

The final experimental M2 bank kept at most one hard distractor per sampled frame and selected up to the three hardest observations. The reliability gate reused Phase 1F geometry criteria:

$$\mathrm{IoU}_{t-1,t} \geq 0.63, \qquad \text{normalized center jump} \leq 0.12.$$

The provisional distractor-opportunity thresholds were similarity ≥ 0.80 and normalized distance ≤ 3.0. These thresholds were used only for this pilot memory study and were not claimed as validated system constants.

The cleaned bank contained negative memory in 5/10 events and 12 total stored entries:

- E003: 3 entries,
- E005: 3 entries,
- E008: 2 entries,
- E009: 3 entries,
- E011: 1 entry.

Events without a hard distractor bank automatically fell back to M1.

6.3 Initial M2 scoring rule

The first controlled M2 hypothesis kept the M1 positive memory unchanged and added only one new factor: negative distractor memory. Candidate scoring was:

$$S_+(c) = \max_{m \in M_+} \cos(c, m), \qquad S_-(c) = \max_{d \in M_-} \cos(c, d),$$

$$S_{M2}(c) = S_+(c) - S_-(c).$$

If no negative memory existed, $S_{M2} = S_+$ and M2 reduced exactly to M1.

Table 3: H-0, M1, and initial M2 ranking performance.

Metric	H-0	M1	M2 subtraction
Rank-1 accuracy	5/10	5/10	4/10
Rank-3 accuracy	6/10	5/10	6/10
Mean target rank	4.70	4.40	4.30
MRR	0.585	0.574	0.551
Positive target margin	5/10	5/10	4/10
Mean target margin	+0.009	+0.0222	-0.0347

Among the five active-negative-memory events, M2 improved two, worsened two, and left one unchanged. Notable event changes were E003: 8→2, E008: 4→9, E009: 1→2, and E011: 6→5.

7 M2 Failure Mechanism and Redesign Lesson

Because M2 did not improve overall performance, Phase 1H did not tune an arbitrary subtraction weight until the pilot results looked better. Instead, a dedicated failure diagnostic compared the true target against the strongest wrong candidate in each active M2 event.

Table 4: M2 failure diagnosis. Positive advantage is target positive evidence minus strongest-wrong positive evidence. Penalty disadvantage is target negative-memory similarity minus strongest-wrong negative-memory similarity.

Event	Positive advantage	Penalty disadvantage	Final advantage
E003	+0.1072	+0.1538	-0.0467
E005	+0.5822	+0.5728	+0.0094
E008	+0.0285	+0.1411	-0.1126
E009	+0.4013	+0.4084	-0.0072
E011	+0.0642	+0.1204	-0.0563

Across all five active events:

- the true target had stronger positive target-memory evidence than the strongest wrong candidate in 5/5;
- the true target also received a larger negative-memory penalty in 5/5;
- after subtraction, the true target still beat the strongest wrong candidate in only 1/5.

This identifies the structural problem: a "hard distractor" is selected precisely because it resembles the target. The true target therefore also resembles the negative-memory bank. Direct subtraction can punish the correct target more than the wrong candidate.

Frozen finding. Distractor memory should not automatically be interpreted as evidence against a candidate. In this project it is better understood as a cue that the appearance decision may be ambiguous and that more identity evidence may be needed.

8 Ambiguity Signals from Candidate Competition

A second diagnostic tested whether the M2 positive-minus-negative ambiguity gap could distinguish correct from wrong M1 Rank-1 choices. It could not: correct and wrong gaps overlapped substantially. Therefore no threshold on that gap was frozen.

The same analysis revealed a more promising runtime signal: the **Top1–Top2 M1 appearance-score margin**. Across all 10 events:

Table 5: All-event M1 Top1–Top2 candidate-margin diagnostic.

Event	Rank-1 correct?	Top1 score	Top1–Top2 margin
E001	No	0.8611	0.0118
E002	Yes	0.8844	0.0442
E003	No	0.8887	0.0203
E004	No	0.8016	0.0836
E005	Yes	0.8783	0.4931
E006	Yes	0.9529	0.0762
E007	Yes	0.9725	0.0988
E008	No	0.8597	0.0072
E009	Yes	0.8929	0.0552
E011	No	0.8756	0.0249

Four of the five wrong Rank-1 choices had very small margins (0.0072–0.0249), while the minimum correct margin was 0.0442. E004 was the exception: it was **confidently wrong** with margin 0.0836. This motivated an event-specific diagnosis rather than prematurely choosing a margin threshold.

9 E004 Confidently-Wrong Failure Analysis

E004 was critical because it showed why margin alone cannot define identity confidence. At frame 720, the true target was present in the Phase 1G candidate set but ranked only 12th by M1 with similarity approximately 0.484. The appearance Rank-1 candidate was a different vehicle with similarity 0.8016. M1 memory size was only 1, so E004 effectively used the original selection appearance only.

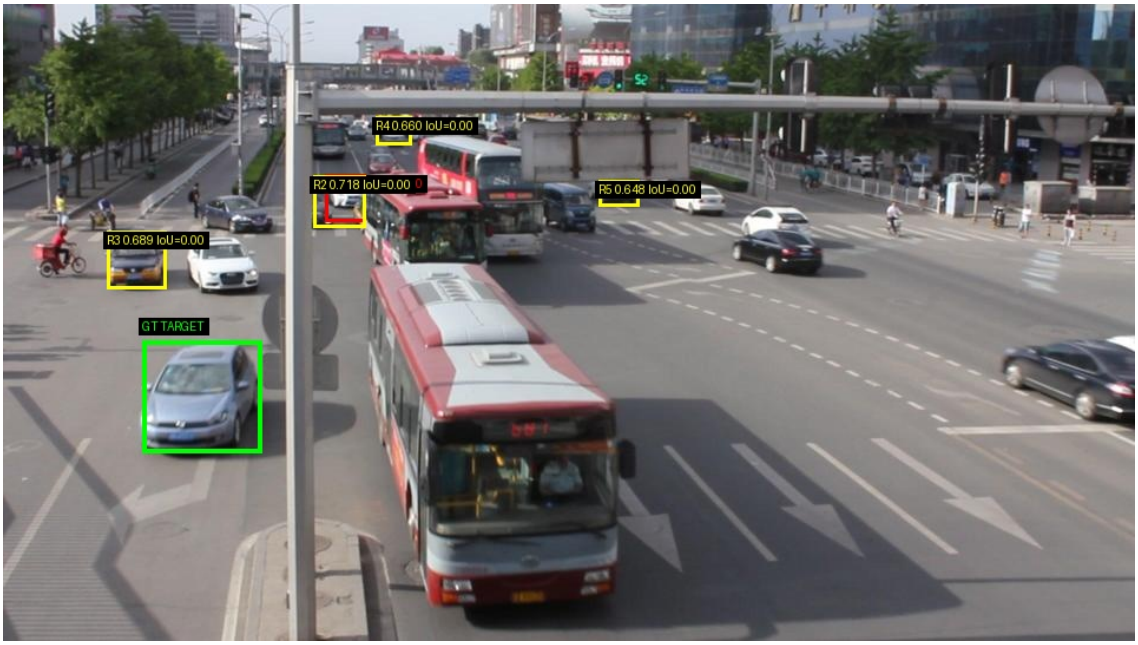


Figure 2: E004 reappearance frame. The true target is shown in green. The Top-5 M1 candidates are red/yellow and all are wrong. The real target was retained by Phase 1G but ranked much lower by appearance.

The visual audit showed two points:

1. The main error is an identity-selection failure (F5): the target exists in the candidate set but the appearance representation does not select it.
2. Rank-1 and Rank-2 are overlapping boxes on essentially the same physical vehicle, but removing these boxes would not solve E004 because the true target remains far below them.

E004 therefore represents a confidently wrong appearance decision, likely driven by strong appearance/scale/viewpoint change together with limited target-view memory, not simply by candidate competition.

10 H-2 Candidate Overlap / Duplicate Diagnostic

Because E004 showed overlapping top-ranked boxes, H-2 measured pairwise candidate IoU across all 10 reappearance events before adding any new deduplication rule.

Table 6: Candidate-overlap prevalence across the 10 eligible events.

Pairwise IoU threshold	Events with at least one pair	Total overlapping pairs
≥ 0.50	6/10	11
≥ 0.70	1/10	1
≥ 0.90	1/10	1

E004's most-overlapping candidate pair had IoU 0.693, just below the strong-duplicate threshold. Moderate overlap is common and can represent real adjacent/occluding vehicles. Strong

duplicates were rare. Since Phase 1G already retained the target in 10/10 eligible events, Phase 1H did **not** add an extra NMS/deduplication rule.

Frozen finding. Candidate duplication was not established as the main identity bottleneck. Phase 1G candidate generation remained frozen.

11 H-3 Observation-Quality Diagnostics

H-3 tested whether appearance decisions should be conditioned on observation quality rather than trusted uniformly.

11.1 True-target quality diagnostic

For analysis only, the actual target candidate was examined after ranking. The clearest separation was area/scale change:

- correct Rank-1 events: mean target $|\log(\text{area ratio})| = 0.2234$,
- wrong Rank-1 events: mean target $|\log(\text{area ratio})| = 1.2256$.

E003 and E004 were extreme, with target areas approximately 13.07x and 5.91x the original selection area. However, this is not itself a deployable routing signal because runtime does not know which candidate is the true target.

11.2 Runtime-visible Top1 signals

When the analysis was restricted to the actually selected Rank-1 candidate, area/aspect changes overlapped heavily between correct and wrong decisions. The strongest runtime-visible signals were instead:

1. Top1–Top2 appearance margin (candidate competition), and
2. detector confidence as an observation-quality signal, especially its rank/percentile within the same frame.

Detector confidence was explicitly interpreted as **observation reliability**, not identity confidence.

11.3 Appearance versus detector-confidence consistency

Table 7: Runtime-visible confidence consistency for M1 Rank-1 candidates.

Event	Correct?	Appearance margin	YOLO conf.	Conf. percentile
E001	No	0.0118	0.461	0.286
E002	Yes	0.0442	0.752	0.786
E003	No	0.0203	0.696	0.667
E004	No	0.0836	0.215	0.118
E005	Yes	0.4931	0.584	0.600
E006	Yes	0.0762	0.636	0.600
E007	Yes	0.0988	0.716	0.900
E008	No	0.0072	0.754	0.643
E009	Yes	0.0552	0.671	0.696
E011	No	0.0249	0.534	0.667

E004 became diagnostically clear: appearance margin 0.0836 looked confident, but the winning candidate had YOLO confidence 0.215, rank 16/18 by detector confidence, confidence percentile 0.118, and a 0.678 gap to the strongest detection in the frame. Thus appearance and observation quality strongly contradicted each other.

Across the pilot set, correct Rank-1 decisions had mean confidence percentile 0.7163 versus 0.4759 for wrong decisions. The ranges still overlap, so no general detector-confidence threshold was claimed.

12 H-3 R1 Versus Exploratory R2 Selective Routing

The final Phase 1H experiment compared two decision behaviors without retraining or reranking candidates.

R1 - always accept appearance Rank-1. Every event outputs MATCH.

Exploratory R2 - selective routing. Accept Rank-1 only when two runtime-visible signals are sufficiently reliable; otherwise output NEED MORE EVIDENCE. For this pilot diagnostic only:

$$\text{appearance margin} \geq 0.03 \quad \text{and} \quad \text{confidence percentile} \geq 0.50.$$

These thresholds were derived from the same 10-event pilot analysis and therefore are **not validated final thresholds**. They were used only to test whether the selective-routing concept is worth carrying forward.

Table 8: R1 versus exploratory R2 selective decision behavior.

Metric	R1	Exploratory R2
Accepted MATCH decisions	10/10	5/10
Correct accepted matches	5/10	5/5
False accepted matches	5/10	0/5
Accepted-match accuracy	0.500	1.000
Coverage	1.000	0.500
NEED MORE EVIDENCE rate	0/10	5/10
False MATCH rate over all events	5/10	0/10

The event-level routing behavior was:

- E001: NEED MORE EVIDENCE - low candidate margin + low observation reliability.
- E003: NEED MORE EVIDENCE - low candidate margin.
- E004: NEED MORE EVIDENCE - low observation reliability despite a large appearance margin.
- E008: NEED MORE EVIDENCE - low candidate margin.
- E011: NEED MORE EVIDENCE - low candidate margin.
- E002, E005, E006, E007, E009: MATCH.

Scientific caution. The 1.000 selective accuracy is a pilot in-sample result, not a generalization claim. The thresholds were selected after observing these same events. The valid conclusion is that selective routing is promising and should be validated on additional events/data before any final operating point is frozen.

13 Consolidated Phase 1H Findings

Phase 1H produced a coherent sequence of findings rather than a single "best model":

1. **Candidate availability is not enough.** Phase 1G retained the target in 10/10 eligible events, yet H-0 appearance ranking achieved only 5/10 Rank-1.
2. **Recent positive target memory was not consistently helpful.** Corrected M1 remained 5/10 Rank-1 and slightly reduced MRR.
3. **Naive negative distractor subtraction was structurally harmful.** The true target received larger negative-memory penalties in all 5 active events.
4. **Distractor memory is better interpreted as ambiguity context than as direct anti-target evidence.**
5. **Candidate-margin ambiguity caught most failures, but not confidently-wrong E004.**
6. **Candidate duplication was not the main bottleneck.** Strong duplicates were rare.

7. **Observation quality matters.** E004 was appearance-confident but detector-quality-poor, motivating reliability-aware routing.
8. **Forcing identity decisions is unsafe.** Exploratory selective routing converted all five pilot false matches into deferrals at 50% coverage, while preserving all five correct accepted matches.

The central research lesson is therefore:

> **Appearance should be treated as one piece of identity evidence, not an obligation to decide. When candidate competition or observation quality makes appearance unreliable, the correct action is to preserve uncertainty and acquire additional evidence.**

This is the first Phase 1 result that directly operationalizes the project's intended MATCH / REJECT / NEED MORE EVIDENCE philosophy.

14 Scientific Cautions and Limitations

The following cautions must be preserved in future paper writing.

14.1 Pilot-scale evidence

All numerical findings are based on 10 eligible UA-DETRAC pilot events. They demonstrate failure mechanisms and justify the next methodology steps; they do not establish population-level performance.

14.2 No independent threshold validation

The exploratory R2 thresholds (margin 0.03, confidence percentile 0.50) were derived after inspecting this pilot set. Therefore the observed 5/5 accepted accuracy and 0/10 false-MATCH rate must not be reported as validated generalization performance.

14.3 Detector confidence is not identity confidence

YOLO confidence is used only as a proxy for observation/detection quality. It does not measure whether a candidate is the same vehicle as the user-selected target.

14.4 Controlled memory cutoff is not a deployable oracle

M1/M2 use the known pre-occlusion interval for a controlled comparison. A deployed online memory policy cannot depend on manual `occlusion_start`; it should use runtime reliability states and suppress updates during persistent CRITICAL/LOST behavior.

14.5 M2 thresholds are provisional

The 0.80 appearance-similarity and 3.0 normalized-distance criteria defined diagnostic hard-distractor encounters. They were not calibrated as final memory-policy parameters.

14.6 No additional cue acquisition yet

Phase 1H ends at MATCH versus NEED MORE EVIDENCE. It does not yet define which cue should be acquired next, how acquisition cost is scored, or when REJECT should be emitted. Those belong to later phases of the sequential acquisition methodology.

14.7 E004 remains an important stress case

E004 shows that an appearance model can be confidently wrong. Any future policy that uses only score margin can still fail on such cases; observation reliability and/or additional identity cues are required.

15 Phase 1H Freeze Decision

Phase 1H is frozen with the following decisions:

1. H-0 ResNet50 cosine ranking remains the simple appearance identity baseline.
2. The square-pad-then-resize preprocessing is frozen for this baseline.
3. Corrected M1 uses original selection + recent clean pre-occlusion target views for controlled evaluation; the leaked reappearance-memory result is discarded.
4. M1 is retained as an experimental baseline but is not claimed to improve H-0 consistently.
5. M2 direct subtraction of distractor similarity is rejected as the final identity scoring rule.
6. Distractor memory is retained conceptually as ambiguity context, not direct negative identity evidence.
7. Phase 1G candidate generation remains unchanged; no extra deduplication/NMS stage is added from H-2.
8. Candidate Top1–Top2 margin and observation reliability are retained as promising selective-decision signals.
9. Exploratory R2 demonstrates the value of MATCH versus NEED MORE EVIDENCE behavior, but its numerical thresholds remain explicitly non-validated and non-final.
10. Phase 1H therefore freezes the **decision architecture**, not a universal final threshold set.

16 Handoff to Phase 1I

Phase 1I should now systematize the failure taxonomy across the frozen pilot events rather than continuing to tune Phase 1H on the same 10 cases. The working taxonomy is:

- F1 - drift before occlusion,
- F2 - target completely lost,
- F3 - tracker jumped to a visually similar distractor,
- F4 - true target returned but was not detected / retained,

- F5 - target was detected but identity ranking selected another candidate,
- F6 - memory contaminated,
- F7 - multiple candidates genuinely ambiguous,
- F8 - target never returned.

The purpose of Phase 1I is to identify which failure modes dominate and, crucially, what additional evidence would resolve them. That analysis will determine which cue should be acquired after NEED MORE EVIDENCE and will prepare the project for the later reliability-aware sequential acquisition policy.

Frozen finding. Phase 1H achieved a concrete research transition: from forced single-cue identity ranking to reliability-aware selective identity decisions. The next methodology step is no longer "improve the appearance ranker at all costs," but "understand unresolved failure modes and acquire additional evidence only when appearance is insufficient."

Archival Note

This document is intended as the source record for later full-paper writing. Frozen findings and cautions here should take precedence over provisional Phase 1H notes. In particular: the leaked early M1 10/10 result is invalid and discarded; M2 subtraction is an analyzed failure, not a final method; and the exploratory R2 100% selective accuracy on accepted pilot decisions must not be presented as an independently validated result.